

Reasoning Over Time – filtering, state estimation, HMM/Kalman-filter intuition

CS 4880: Artificial Intelligence

Lecture 21 – October 14, 2026

Kirk Duran | Cal Poly, San Luis Obispo

Today

- ▶ Course unit: Course sequence
- ▶ Big idea: Reasoning Over Time – filtering, state estimation, HMM/Kalman-filter intuition
- ▶ Placeholder for announcements and context.

Learning Goals

- ▶ Define the key vocabulary for this lecture.
- ▶ Work through one representative example.
- ▶ Connect today's idea to the previous and next class meetings.

Placeholder

Replace these bullets with the final lecture material, examples, and in-class activities.

- ▶ Main concept 1.
- ▶ Main concept 2.
- ▶ Main concept 3.

Example or Activity

In class

Add a short worked example, code trace, proof sketch, discussion prompt, or board activity here.

Wrap-Up

- ▶ What should students be able to do after today?
- ▶ What should they practice before the next meeting?
- ▶ What question should they bring to class next time?